

Problem 3

Determine the real and imaginary parts of the complex numbers:

$$1^\circ -7i; \quad 2^\circ 2-4i; \quad 3^\circ \cos \alpha + i \sin \alpha - 1; \quad 4^\circ a + ib + c + i$$

Solution

For each complex number, we will write it in the standard form $z = x + iy$ and identify the real part $\operatorname{Re}(z) = x$ and imaginary part $\operatorname{Im}(z) = y$.

Recall: Standard Form

A complex number in standard form is written as:

$$z = x + iy$$

where:

- $x = \operatorname{Re}(z)$ is the real part
- $y = \operatorname{Im}(z)$ is the imaginary part
- Both x and y are real numbers
- i is the imaginary unit with $i^2 = -1$

Important: The imaginary part is the coefficient of i , not including i itself.

Part 1°: $z = -7i$

Standard Form

$$z = -7i = 0 + i \cdot (-7)$$

Real and Imaginary Parts

$\operatorname{Re}(z) = 0, \quad \operatorname{Im}(z) = -7$

This is a purely imaginary number on the negative imaginary axis.

Part 2°: $z = 2 - 4i$

Standard Form

$$z = 2 - 4i = 2 + i \cdot (-4)$$

Real and Imaginary Parts

$\operatorname{Re}(z) = 2, \quad \operatorname{Im}(z) = -4$

This complex number has both a real part and an imaginary part.

Part 3°: $z = \cos \alpha + i \sin \alpha - 1$

Rearrange to Standard Form

First, group the real and imaginary terms:

$$z = \cos \alpha + i \sin \alpha - 1$$

$$= (\cos \alpha - 1) + i \sin \alpha$$

Real and Imaginary Parts

$\operatorname{Re}(z) = \cos \alpha - 1, \quad \operatorname{Im}(z) = \sin \alpha$
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Note: Both parts depend on the parameter α .

Special Cases

- If $\alpha = 0$: $\operatorname{Re}(z) = 1 - 1 = 0$, $\operatorname{Im}(z) = 0$, so $z = 0$
- If $\alpha = \frac{\pi}{2}$: $\operatorname{Re}(z) = 0 - 1 = -1$, $\operatorname{Im}(z) = 1$, so $z = -1 + i$
- If $\alpha = \pi$: $\operatorname{Re}(z) = -1 - 1 = -2$, $\operatorname{Im}(z) = 0$, so $z = -2$

Part 4°: $z = a + ib + c + i$

Rearrange to Standard Form

Group all real terms together and all imaginary terms together:

$$z = a + ib + c + i$$

$$= (a + c) + i(b + 1)$$

Here we assume a, b, c are real numbers (constants or parameters).

Real and Imaginary Parts

$\operatorname{Re}(z) = a + c, \quad \operatorname{Im}(z) = b + 1$
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Note: Both parts depend on the parameters a , b , and c .

Verification

We can verify: $(a + c) + i(b + 1) = a + c + ib + i = a + ib + c + i$

Summary Table

Complex Number z	Real Part $\operatorname{Re}(z)$	Imaginary Part $\operatorname{Im}(z)$
$-7i$	0	-7
$2 - 4i$	2	-4
$\cos \alpha + i \sin \alpha - 1$	$\cos \alpha - 1$	$\sin \alpha$
$a + ib + c + i$	$a + c$	$b + 1$

Key Points to Remember

1. The imaginary part is a **real number** (the coefficient of i), not including i itself.
2. To find real and imaginary parts:
 - Group all terms without i (these form the real part)
 - Group all coefficients of i (these form the imaginary part)
3. For $z = x + iy$:
 - $\operatorname{Re}(z) = x$
 - $\operatorname{Im}(z) = y$ (NOT iy)
4. Both real and imaginary parts can be functions of parameters (like α, a, b, c).

Additional Example

For clarity, consider $z = 3 + 5i$:

- **Correct:** $\operatorname{Re}(z) = 3, \operatorname{Im}(z) = 5$
- **Incorrect:** $\operatorname{Im}(z) = 5i$ (the imaginary part is just the coefficient)

Final Answer

1°	$z = -7i :$	$\operatorname{Re}(z) = 0,$	$\operatorname{Im}(z) = -7$
2°	$z = 2 - 4i :$	$\operatorname{Re}(z) = 2,$	$\operatorname{Im}(z) = -4$
3°	$z = \cos \alpha + i \sin \alpha - 1 :$	$\operatorname{Re}(z) = \cos \alpha - 1,$	$\operatorname{Im}(z) = \sin \alpha$
4°	$z = a + ib + c + i :$	$\operatorname{Re}(z) = a + c,$	$\operatorname{Im}(z) = b + 1$

General Principle: To find the real and imaginary parts of a complex number:

1. Write the number in standard form $z = x + iy$
2. The real part is the sum of all terms without i
3. The imaginary part is the sum of all coefficients of i